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Short Communication

Improving eye care follow-up adherence in diabetic patients with ocular abnormalities: the effectiveness of patient contracts in a free, pharmacy-based eye screening

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ABSTRACT

Objectives: Patient contracts are increasingly utilized in medical practice and have the potential to improve health outcomes in high-risk populations. However, as a relatively new tool, there has been limited research regarding the efficacy of patient contracts. Diabetic retinopathy (DR) is one of the leading causes of vision impairment in adults in the US and only 50–60% of adults with diabetes adhere to annual dilated fundus exam recommendations. This study aimed to evaluate the impact of patient contracts on follow-up adherence in diabetic patients with ocular abnormalities after a free, pharmacy-based eye screening.

Study design: This prospective study implemented a non-invasive, non-mydriatic fundus camera in an urban, community-based pharmacy setting to screen for ocular diseases in patients with diabetes. Patients were assigned to the contract or non-contract group. Patients who signed a contract agreed to: 1) review their results with their primary care doctor, 2) follow-up with an ophthalmologist if their results were abnormal, and 3) inform research staff if/when they completed an eye care appointment. All study participants and their primary care doctors were notified of their results via mail. Follow-up questionnaires were administered to all patients by telephone three months after the screening results.

Results: 500 patients were screened and 113 (22.6%) had abnormal results. Of the patients who had abnormal results, 83 (74.3%) were able to be contacted. Of the 83 patients who were able to be contacted, the majority of patients were African American (73.5%) and female (56.6%). The mean age was 54.7 years. Of those, 34 (41.0%) adhered to follow-up recommendations. There was no significant difference in follow-up adherence between the contract (38.1%) and non-contract group (43.9%) ($P = 0.59$). In addition, 70.4% of patients did not comply with at least one measure of the contract agreement.

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Conclusion: Contracts did not increase follow-up adherence to eye appointments in diabetic patients with ocular abnormalities. The majority of patients did not comply with their contract and follow-up adherence was low in both groups. Most research has yielded mixed results regarding the efficacy of contracts in improving health outcomes. Therefore, different types of contracts or other patient-centered tools should be evaluated in order to increase follow-up adherence in patients at high risk for DR.
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Introduction

Patient contracts are increasingly utilized in medical practice, with specific uses ranging from preventing drug abuse to promoting weight loss. In these contexts, a patient contract refers to an agreement in which a patient contractually obliges to commit to a set of healthy behaviours.[1] As a patient-centered tool, contracts can facilitate communication and help patients understand physicians' expectations. If effective, contracts have the potential to improve health outcomes.

Though patient contracts have been studied for over two decades, there is limited research regarding their efficacy. Most research to date has involved small samples and yielded mixed results.[1–6] In a meta-analysis of 30 trials assessing the impact of patient contracts, the outcome measure favoured contracts in 15 trials, the outcome measure favoured the control group in six trials, and an outcome measure with no differences between groups was observed in 26 trials. The medium sample size in these 30 trials was 21 subjects, with only two studies having sample sizes over 100.[1]

Because of mixed results in the literature and confounding variables, more research is needed to determine the efficacy of patient contracts, especially in the field of eye care. Only one trial, related to contact lens hygiene, examined contract efficacy in eye care and the results were inconclusive due to high adherence in both groups.[2] If effective, patient contracts could be used to target high-risk populations who exhibit low adherence to follow-up eye care. Therefore, contracts may be especially beneficial to patients who are at risk for developing diabetic retinopathy (DR). DR is one of the leading causes of vision impairment in adults in the US. Currently, the American Diabetes Association recommends that all people with diabetes have annual dilated fundus examinations (DFEs) to reduce their risk of vision loss. However, only 50% of older people with diabetes follow this recommendation.[7]

This prospective study implemented a contract intervention in diabetic patients who received a free, non-dilated fundus exam in an urban, community-based pharmacy. We aimed to assess the impact of contracts on follow-up adherence in participants with observable ocular abnormalities who were recommended to follow-up with an eye care provider.

Recruitment and contract terms

Following Institutional Review Board approval, participants were recruited from an urban outpatient pharmacy in Philadelphia, PA. Participants had to be at least 18 years of age and diagnosed with diabetes (either self-reported or identified by a pharmacist through medication prescriptions). English and Spanish-speaking participants were eligible.

Following informed consent, participants were assigned to the contract or non-contract group. Participants who were assigned to the contract group signed a contract in which they agreed to: 1) review their examination results with their primary care doctor, 2) follow-up with an ophthalmologist if recommended, and 3) complete a 3-month follow-up questionnaire administered by the research staff via telephone. The contracts were explained to each participant and were emphasized to not have any legal impact on their health care.

Delivery of screening results and follow-up questionnaire administration

Participants were notified of whether their screening results were normal or abnormal via mail within three weeks of their screening. An abnormal result meant that an ocular abnormality was detected, indicating possible DR or another ocular disease. All participants with abnormal results were instructed to follow-up with an ophthalmologist within a specified time frame, depending on the diagnosis and the severity of disease. Within one month of receiving their results, participants with abnormal findings, regardless of being in the contract or non-contract group, also received a phone reminder advising them to make an appointment with an ophthalmologist.

Research assistants administered a follow-up questionnaire to all participants with abnormal findings by telephone three months after the screening results were distributed. The questionnaire addressed follow-up eye care utilization, satisfaction with the screening, and barriers to obtaining eye care.

A Chi-squared test was used to examine the association between contracts and follow-up adherence in participants with abnormal results. Statistical analyses were performed using SAS V9.3 software (SAS Inc., Cary, NC).

Screening results and contract adherence

In total, 500 participants received fundus exams at the outpatient pharmacy location. The participants were assigned to groups by alternating weeks; the 250 patients who attended screenings during odd weeks were assigned to the contract group and the 250 patients who attended screenings during even weeks were assigned to the non-contract group.

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Of the 500 participants screened, 113 (22.6%) were found to have DR or another ocular abnormality and were advised to follow-up with an ophthalmologist within a recommended time frame. Of the 113 participants who had abnormal results, 83 (74.3%) were able to be contacted and complete the 3-month follow-up questionnaire regarding their follow-up eye care utilization. Of the 83 participants who could be contacted, the majority of participants were African American (73.5%) and female (56.6%). The mean age was 54.7 years. There were no significant differences in the demographic composition of the contract and non-contract groups. Table 1 presents the percentage of participants who adhered to the recommended follow-up guidelines, which was defined as scheduling and keeping an appointment with an ophthalmologist. In total, only 34 (41.0%) participants reported that they followed up with an ophthalmologist. There was no significant difference between follow-up adherence in the contract group (38.1%) and the non-contract group (43.9%).

Given that one of the terms of the contract was to complete the 3-month follow-up questionnaire, unreachable participants breeched the conditions of their contract. Overall contract adherence was poor, with 70.4% not complying with at least one condition of their contract agreements.

Impact on follow-up eye appointment adherence

A contract did not increase follow-up eye appointment adherence in participants with ocular abnormalities after a free, pharmacy-based eye screening.

Overall, follow-up adherence in participants with abnormal screening results was poor in both groups (38.1% for the contract group and 43.9% for the non-contract group). Contract utilization did not have a significant effect on adherence and the majority of participants in the contract group breeched at least one condition of their contract (70.4%). Despite low follow-up adherence and contract adherence rates, the majority of participants were satisfied with their screening (96.8%) and reported no barriers to obtaining eye care (70.9%). Patients who reported barriers were less likely to adhere to follow-up recommendations but the effect was not significant (OR = 0.47, *P*-value = 0.19).

Possible explanations for low contract adherence

There are several possible explanations for low contract adherence. In the follow-up survey, 29.1% of patients reported experiencing barriers to obtaining eye care. These barriers likely made it more difficult for patients to adhere to the contract. Frequently cited barriers to care include transportation, insurance, cost and lack of disease education.[8] In this study, cost was the most frequently cited barrier to care. Lack of education about DR is another barrier that could have greatly affected adherence rates. If patients were unaware of DR's severe prognosis and had not yet experienced vision changes because the disease is asymptomatic in early stages, then they potentially would not have understood the purpose of following up with an ophthalmologist.[8]

Most research to date has focused on patient contracts administered by physicians.[1] Several studies have used other medical staff members, family members, or peers to administer contracts.[9] However, there is little data noting outcomes when research assistants administered contracts. This may have an impact on contract adherence. Previous studies have shown that trust between parties is important for contracts to be efficacious and that participants who have an established relationship with the person administering the contract may feel more of an obligation to abide by the agreement.[10] In addition, participants were recruited from an outpatient setting and were not necessarily already seeing an eye care provider. Therefore, they may not have felt an obligation to contact a new physician with whom they had no prior relationship and who did not know about the contract. As such, contracts may prove more effective if administered by someone that the patient has a previous relationship with, such as their eye care provider. There was no educational component or financial incentive in this study. These elements may have been particularly beneficial in our study, as we targeted an urban, underserved population. Other contract studies have shown that financial rewards have been effective in lower socio-economic populations.[1] Finally, none of the terms in our contract resulted in an immediate, observable benefit to participants. Most effective contract studies have involved direct, observable benefits, such as weight loss.[1] In this study, abiding by the terms of the contract would most likely not result in an immediate benefit, even if long-term vision loss would be prevented. A combination of the above factors may have resulted in an overall lack of motivation or ability to abide by the terms of the contract.

Table 1 – Follow-up adherence in contract and non-contract groups.

	Contract ($n = 42$)		Non-contract ($n = 41$)		*P*-value
	n	%	*n*	%	
Follow-up appointment					
Adherent	16	38.1	18	43.9	0.59
Non-adherent	26	61.9	23	56.1	

The need for patient-centered tools in patients at high-risk for DR

In conclusion, patient contracts implemented in a community setting were not effective in increasing adherence to follow-up eye care appointments. Further research on the efficacy of contracts with educational or financial components, or other patient-centered tools may provide methods to increase follow-up adherence in populations at high-risk for DR.

Author statements

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Ethical approval

Institutional Review Board approval was received prior to study implementation. Informed consent was obtained from each participant in accordance with the Declaration of Helsinki.

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Competing interests

None declared.

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